Python Programming Assignment

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Question 11(a): Employee Payroll Preparation (10 Marks)

Employee information is stored in a CSV file, while salary details are maintained in a JSON file. Write a Python program to merge the datasets using Employee ID, identify employees with missing salary records, and prepare a payroll report.

Sample Input: employees.csv

```
EmployeeID,Name,Department
E101,Arjun Nair,Sales
E102,Bhavana Rao,IT
E103,Chandran S,IT
E104,Deepa K,HR
E105,Elango R,Sales
```

Sample Input: salary.json

```
[
    {"EmployeeID": "E101", "Basic": 30000, "HRA": 12000, "Allowance": 3000},
    {"EmployeeID": "E102", "Basic": 45000, "HRA": 18000, "Allowance": 5000},
    {"EmployeeID": "E104", "Basic": 28000, "HRA": 10000, "Allowance": 2000}
]
```

Python Program

```
import csv
import json

# Read employee details from CSV
employees = {}
with open('employees.csv') as f:
    for row in csv.DictReader(f):
        employees[row['EmployeeID']] = row

# Read salary details from JSON
with open('salary.json') as f:
    salary = {s['EmployeeID']: s for s in json.load(f)}

# Merge datasets and find missing salary records
report = ["EMPLOYEE PAYROLL REPORT", f"{'ID':<8}{'Name':<15}{'Dept':<10}{'Gross Pay'}"]
missing = []

for eid, emp in employees.items():
    if eid in salary:
        s = salary[eid]
        gross = s['Basic'] + s['HRA'] + s['Allowance']
        report.append(f"{eid:<8}{emp['Name']:<15}{emp['Department']:<10}{gross}")
    else:
        missing.append(emp['Name'])

report.append("\nEmployees with missing salary records:")
report.append(", ".join(missing) if missing else "None")

report_text = "\n".join(report)
print(report_text)
with open('payroll_report.txt', 'w') as f:
    f.write(report_text)
```

Sample Output

```
EMPLOYEE PAYROLL REPORT
ID      Name           Dept      Gross Pay
E101    Arjun Nair     Sales     45000
E102    Bhavana Rao    IT        68000
E104    Deepa K        HR        40000

Employees with missing salary records:
Chandran S, Elango R
```

Question 11(b): Movie Database Management (10 Marks)

A movie streaming platform stores movie information in XML format and viewer ratings in CSV format. Write a Python program to combine the datasets, calculate the average rating for each movie, and generate a report sorted by rating.

Sample Input: movies.xml

```
<Movies>
    <Movie>
        <MovieID>M1</MovieID>
        <Title>The Last Voyage</Title>
        <Genre>Adventure</Genre>
    </Movie>
    <Movie>
        <MovieID>M2</MovieID>
        <Title>City Lights Return</Title>
        <Genre>Drama</Genre>
    </Movie>
    <Movie>
        <MovieID>M3</MovieID>
        <Title>Silent Horizon</Title>
        <Genre>Thriller</Genre>
    </Movie>
</Movies>
```

Sample Input: ratings.csv

```
RatingID,MovieID,Viewer,Rating
R1,M1,Ravi,4.5
R2,M1,Meena,4.0
R3,M2,Ravi,3.5
R4,M2,Sita,4.2
R5,M3,Meena,5.0
R6,M3,Ravi,4.8
R7,M3,Sita,4.6
```

Python Program

```
import xml.etree.ElementTree as ET
import csv

# Read movie details from XML
tree = ET.parse('movies.xml')
movies = {}
for m in tree.getroot().findall('Movie'):
    mid = m.find('MovieID').text
    movies[mid] = {'Title': m.find('Title').text, 'Genre': m.find('Genre').text}

# Read viewer ratings from CSV
ratings = {mid: [] for mid in movies}
with open('ratings.csv') as f:
    for row in csv.DictReader(f):
        ratings[row['MovieID']].append(float(row['Rating']))
# Calculate average rating for each movie
```

```
avg_ratings = {mid: (sum(r) / len(r) if r else 0) for mid, r in ratings.items()}

# Sort movies by average rating, descending
sorted_movies = sorted(movies.items(), key=lambda x: -avg_ratings[x[0]])

# Generate report
report = ["MOVIE RATING REPORT", f"{'ID':<6}{'Title':<22}{'Genre':<12}{'Avg Rating'}"]
for mid, m in sorted_movies:
    report.append(f"{mid:<6}{m['Title']:<22}{m['Genre']:<12}{avg_ratings[mid]:.2f}")

report_text = "\n".join(report)
print(report_text)
with open('movie_report.txt', 'w') as f:
    f.write(report_text)
```

Sample Output

```
MOVIE RATING REPORT
ID    Title                 Genre       Avg Rating
M3    Silent Horizon        Thriller    4.80
M1    The Last Voyage       Adventure   4.25
M2    City Lights Return    Drama       3.85
```